

COMP1870 Theoretical Foundations of Computer Science I

School of Computer Science, University of Leeds

Worksheet - Week 1.2

This worksheet contains a combination of *formative activities* (which contribute towards your learning) and *summative activities* (which you will complete and submit to be assessed as part of your portfolio).

- Exercises marked with red borders are summative and must be submitted in Gradescope.
- Other exercises may be covered in tutorial sessions.
- Exercises marked by (*) are advanced, and may take some time to complete.
- Exercises marked by (E) are exam-style questions. You should expect to see questions like this in the exam.

Expectations:

- 1. Timeliness** You should complete all of the activities in the order provided and submit your portfolio evidence on PebblePad before the deadline.
- 2. Presentation:** You should present all of your work clearly and concisely following any additional guidance provided by the module staff.
- 3. Integrity:** You are responsible for ensuring that the evidence you submit as part of your portfolio is **entirely your own work**. You can find out more about Academic integrity on the Skill@library website. All work you submit for assessment is subject to the academic integrity policy.

Feedback: Feedback on formative activities will be provided via Lab classes and tutorials. Feedback on evidence submitted as part of the portfolio will be available on Gradescope.

Support opportunities: Support with the activity sheet is available in tutorials and the Math Lunch Hour. Individual support is available via office hours.

Expected completion date: Friday 16 October 2025

All portfolio tasks are rated AMBER by the university: AI tools can be used in an assistive role. You are permitted to use AI tools for specific defined processes within the assessment. Within this assessment you may use Generative AI to:

- help you start to understand more complex ideas by providing accessible summaries.
- test your knowledge against a piece of content.
- help identify and correct spelling mistakes and grammatical errors in your work.
- provide feedback and advice on your overall coding style.

You must not use Gen AI to:

- produce written content (code, comments or any other content) which you then submit as your work, regardless of whether you make changes to it.
- rewrite or make changes to any of your work.

Learning outcomes

On completion of this activity sheet you will have:

1. reasoned about logical expressions using logical equivalencies,
2. describe the strengths and limitations of propositional and predicate logic,
3. translated logical statements in natural language to expressions in propositional and predicate logic.

Tutorials on Wednesday and Thursday will cover material from Week 1.2, Lecture 1. This material is in exercises 1–5. Tutorials on Monday and Tuesday will cover material from Week 1.2, Lecture 2. This material is in exercises 6–14.

Questions

1. (E) In this question, assume that the domain is the set of positive integers and use the following predicates:

$P(x)$ means x is prime

$Q(x)$ means x is a power of a prime number

$S(x)$ means x is a perfect square

$D(x, y)$ means x divides y

$x = y$ means x is equal to y

- (a) Translate the following statement into predicate logic:

There exists a positive integer x such that x is a perfect square and there exist positive integers y and z such that y is a prime and z is a prime and y is not equal to z and y divides x and z divides x .

Is this statement true or false?

- (b) Translate the following statement into predicate logic:

For every positive integer x , if x is a prime power then there exists a positive integer y such that y is prime and for every positive integer z , if z divides x and z is prime then z is equal to y .

Is this statement true or false?

- (c) Translate the following statement into English:

$$\forall x (Q(x) \Rightarrow \forall y \forall z (P(z) \wedge P(y) \wedge D(y, x) \wedge D(z, x) \Rightarrow y = z))$$

Is this statement true or false?

- (d) Translate the following statement into English:

$$\exists x (S(x) \wedge P(x))$$

Is this statement true or false?

2. (E) Use the logical equivalences in Sections 2.1 and 2.2 of the lecture notes to show that

$$\neg \forall x (P(x) \Rightarrow \exists y (P(y) \vee \neg Q(x))) \quad \text{is equivalent to} \quad \exists x (P(x) \wedge \forall y (\neg P(y) \wedge Q(x))).$$

At each step, state which equivalence you are using.

3. Use the logical equivalences in Sections 2.1 and 2.2 of the lecture notes to show that

$$\neg(\exists x \forall y (P(x) \wedge ((Q(x) \vee S(y)) \Rightarrow D(x, y))))$$

is equivalent to

$$\forall x \exists y (\neg P(x) \vee ((Q(x) \vee S(y)) \wedge \neg D(x, y)))$$

At each step, state which equivalence you are using.

4. (E) Use equivalences to find a statement in predicate logic which is equivalent to

$$\neg(\forall x ((\exists y Q(y)) \Rightarrow (\exists z R(z))))$$

where every negation sign appears directly before a predicate.

5. (E) By completing truth tables, decide whether the following arguments are valid.

(a) $P_1 \Rightarrow (P_2 \vee P_3), \neg P_2 \therefore P_1 \Rightarrow P_3$

(b) $P_1 \Rightarrow (P_2 \vee P_3), P_2 \therefore P_1 \Rightarrow P_3$

(c) $P_1 \Rightarrow P_2, P_1 \wedge (\neg P_2) \therefore P_1$

6. (E) The Fitch diagram below shows a proof that $P_1 \wedge (P_2 \vee P_3), (P_2 \vee P_3) \Rightarrow P_4 \therefore P_1 \wedge P_4$ is a valid argument. Fill in all of the justifications on the right hand side.

1. $P_1 \wedge (P_2 \vee P_3)$	_____
2. $(P_2 \vee P_3) \Rightarrow P_4$	_____
3. P_1	_____
4. $P_2 \vee P_3$	_____
5. P_4	_____
6. $P_1 \wedge P_4$	_____

7. The Fitch diagram below shows a proof that $P_1 \Rightarrow (P_2 \wedge P_3), (P_2 \wedge P_4) \Rightarrow P_5, P_1 \wedge P_4 \therefore P_5$ is a valid argument. Fill in all of the justifications on the right hand side.

1. $P_1 \Rightarrow (P_2 \wedge P_3)$	_____
2. $(P_2 \wedge P_4) \Rightarrow P_5$	_____
3. $P_1 \wedge P_4$	_____
4. P_1	_____
5. $P_2 \wedge P_3$	_____
6. P_2	_____
7. P_4	_____
8. $P_2 \wedge P_4$	_____
9. P_5	_____

8. The Fitch diagram below shows a proof that

$$P_1 \vee (P_2 \wedge P_3) \therefore (P_1 \vee P_2) \wedge (P_1 \vee P_3)$$

is a valid argument. Fill in all of the justifications on the right hand side.

1. $P_1 \vee (P_2 \wedge P_3)$	_____
2. P_1	_____
3. $P_1 \vee P_2$	_____
4. $P_1 \vee P_3$	_____
5. $(P_1 \vee P_2) \wedge (P_1 \vee P_3)$	_____
6. $P_1 \Rightarrow ((P_1 \vee P_2) \wedge (P_1 \vee P_3))$	_____
7. $P_2 \wedge P_3$	_____
8. P_2	_____
9. P_3	_____
10. $P_1 \vee P_2$	_____
11. $P_1 \vee P_3$	_____
12. $(P_1 \vee P_2) \wedge (P_1 \vee P_3)$	_____
13. $(P_2 \wedge P_3) \Rightarrow ((P_1 \vee P_2) \wedge (P_1 \vee P_3))$	_____
14. $(P_1 \vee P_2) \wedge (P_1 \vee P_3)$	_____

9. The Fitch diagram below shows a proof that $P_1 \Rightarrow P_2 \therefore (\neg P_1) \vee P_2$ is a valid argument. Fill in all of the justifications on the right hand side.

1. $P_1 \Rightarrow P_2$	_____
2. $\neg((\neg P_1) \vee P_2)$	_____
3. P_1	_____
4. P_2	_____
5. $(\neg P_1) \vee P_2$	_____
6. $((\neg P_1) \vee P_2) \wedge \neg((\neg P_1) \vee P_2)$	_____
7. $P_1 \Rightarrow ((\neg P_1) \vee P_2) \wedge \neg((\neg P_1) \vee P_2)$	_____
8. $\neg P_1$	_____
9. $(\neg P_1) \vee P_2$	_____
10. $((\neg P_1) \vee P_2) \wedge \neg((\neg P_1) \vee P_2)$	_____
11. $\neg((\neg P_1) \vee P_2) \Rightarrow ((\neg P_1) \vee P_2) \wedge \neg((\neg P_1) \vee P_2)$	_____
12. $\neg\neg((\neg P_1) \vee P_2)$	_____
13. $(\neg P_1) \vee P_2$	_____

10. Write a Fitch diagram which shows that

$$\neg(P_1 \wedge P_2) \therefore P_1 \Rightarrow (\neg P_2)$$

is a valid argument. (It is possible to do this in 8 lines.)

11. Write a Fitch diagram which shows that

$$\neg(P_1 \vee P_2) \therefore (\neg P_1) \wedge (\neg P_2)$$

is a valid argument. (It is possible to do this in 12 lines.)

12. The following Fitch diagram shows a “proof” that the argument $P_1 \Rightarrow P_2 \therefore \neg P_1$ is a valid argument.

- (a) Verify that $P_1 \Rightarrow P_2 \therefore \neg P_1$ is not a valid argument by constructing truth tables.
 (b) On what line of this Fitch diagram does an error occur? What is the error?

1. $P_1 \Rightarrow P_2$	Premise
2. P_1	Assumption
3. P_2	$\Rightarrow$ Elim 1, 2
4. $\neg P_2$	$\neg$ Intro 3
5. $P_2 \wedge (\neg P_2)$	$\wedge$ Intro 3, 4
6. $P_1 \Rightarrow (P_2 \wedge (\neg P_2))$	$\Rightarrow$ Intro 2-5
7. $\neg P_1$	$\neg$ Intro 6

13. (*) Use the rules of inference to prove that

$$(\neg P_1) \wedge (\neg P_2) \therefore \neg(P_1 \vee P_2)$$

is a valid argument. It is possible to do this in 18 lines. (Can you find a way to do it in fewer than 18?)

14. (*) Use the rules of inference to prove that

$$\neg(P_1 \wedge P_2) \therefore (\neg P_1) \vee (\neg P_2)$$

is a valid argument. It is possible to do this in 24 lines. (Can you find a way to do it in fewer than 24?)